Section T (v3.1). Alias phase sums with ternary left masks (Theorem A′)

Companion file to J. Voitle, "A certified exponent of 0.48758 for worst-case Subset Sum, and limits of the alias–checksum analysis" (draft, 5 November 2026). It gives the complete proofs of the ternary alias step stated in Appendix A of the paper, behind Theorem 3 (conditional on [S] and on Base's written proof). File names cited below refer to the project repository and its verification snapshot.

v3.1 (5 November 2026): (EM), (SP) and (SK) are stated with the explicit ε0 slack; no number changes. The historical candidate hash below identifies the original JSON, including its metadata. The accompanying snapshot preserves every numerical input and corrects or omits obsolete display metadata; use MANIFEST.json for its file hashes.

v3 (round 7): applies the repairs of Astra's fresh referee report (`REFEREE_SECTION_T.md`, verdict PASS WITH REPAIRS). All changes are to the text; no number changes. The frozen point `candidate_ternary.json` (sha256 ea2f10e89ea5f0073479e1e41cc5e22ffda31a5655966ee3f458e8cff25c422b) and all 33 certified conditions are unchanged (`scripts/round7/out/recheck_v3.txt`). The changes: Base's scope (word-RAM model, success ≥ 2/3, b ≤ n^c) is restored in the statement; the trial count is an explicit integer; the trial bound in Lemma T7 carries its O(log n); the parameter table lists every condition Astra's inventory derives, including the small-prime and right-key-sort totals and the cap margin; Lemmas T5–T6 are stated for the uncapped ideal execution; Lemma B2's a_Γ ≤ ¾ is explained; the cutoffs are made explicit; cosmetic items. The diff is listed in `CHANGELOG_T_v2_to_v3.md`.

v2 (round 3): [S] carries a polynomial constant C_S(m) (Conjecture M′ + Lemma S); the remark "C_S = 3 if [S] comes from monotonicity" is withdrawn (exact monotonicity is false, ROUND2_REPORT A.3).

Written in the style of Base (`base_PROOF_0.4877.tex`, cited by line number as "Base l.…"). It replaces Base's Section 5 (l.301–612). Everything not restated is as in Base. The result is **conditional on [S]** (Section A of `ROUND2_REPORT.md`: a_{k′}(M; w) ≥ 2^{−d(M)}/C_S(m) for every signing w, k′ ≤ 0.11m, with C_S(m) a fixed polynomial; under Conjecture M′ of ROUND2_REPORT A.6, C_S(m) = P(m) + 2 by Lemma S) **and on Base's written proof**; the checksum (Base Section 6 with J = 1, as in paper_v2 Section 8) is unchanged.

T.1 Parameters and constraints

Parameters τ0, η, μ, γ, l as in Base l.136–156, with r = 1 − μ − l, Δ = η − τ0, Γ = 2τ0 − η, a_Γ = Δ/Γ, ε = ζ = 10^{-8}, guard $10^8\lceil\log(b+1)\rceil \le n$ (Base l.193–194 with $10^6 \to 10^8$; it implies b + 2 ≤ 2^{n/10^8+1}). New: two mask fractions a (inside) and c (outside), with a′ := a + c. Sizes: m = 2⌊μn/2⌋, σ = m/2, k1 = ⌊am⌋, k2 = ⌊cm⌋, k′ = k1 + k2, |L| = ⌊ln⌋, |R| = n − m − |L|. Write H3(a,c) = −a log a − c log c − (1−a−c) log(1−a−c) (so the number of left masks is 2^{H3(a,c)m + O(log n)}) and h for binary entropy.

Constraints. All are certified in `scripts/round2/check_ternary_point.py` (arb, 200 bits) and, independently, in Astra's `verify_a.py` (`SCALAR_SLACKS.md`, 50 decimal digits). The inequalities marked

"<" are strict; (MU2), (SR), the domain condition $2a' \le \tfrac{1}{2}$ of (A4R), $a_\Gamma \le \tfrac{3}{4}$ and $p, q \ge 1$ are non-strict by nature. At the certified point every condition has positive slack, so the distinction does not matter there. Work bounds are proved in Lemma T7 ("T7 item k" below); coverage in Lemma T5.

	Inequality	**Used in**
(C1R)	$\gamma < h(2a)/2 + h(2c)/2$	T5 step 4: $2^{\gamma m} \le C(\sigma,k1)C(\sigma,k2)$ for large n
(C2R)	$l + (H3(a,c) - \gamma)\mu < \tau 0$	T7 item 5: left records
(C3)	$r + (1 - \gamma)\mu < \tau 0$	T7 item 6: right class records (Base l.167)
(C4R)	$1 - \mu + (H3(a,c) + h(\tfrac{1}{2} - a + c) - \gamma)\mu - \eta < \tau 0$	T7 item 7: visited tuples
(C5R)	$\max(H3(a,c), h(\tfrac{1}{2} - a + c))\mu + \tfrac{1}{2} - \tau 0 < \tau 0$	T7 item 2: both mask lists per trial $\times$ trials
(C6)	$0 < r < \tau 0$, $l < \tau 0$	T7 item 1: value maps (Base l.170)
(MU1)	$\mu < l$	T3/T7: $d(L) \ge d(M) - 2n^{0.6}$ w.h.p. (replaces Base's (C7))
(MU2)	$\mu \le \tfrac{1}{2}$	T3: $d(M) \le f(\lfloor n/2 \rfloor) + n^{0.6}$ under (LD) (replaces (S2))
(A4R)	$\mu(h(2a') + 2a') < \eta$, with the domain condition $2a' \le \tfrac{1}{2}$	T4: all $\le 2k'$-sparse relations survive reduction mod P
(SR)	$a' \le 0.11$	range of [S] as derived from Conjecture M′ (Lemma S needs $h(a') \le \tfrac{1}{2}$); not a separate premise when [S] is assumed directly
(EM)	$\mu < \tau 0 - \varepsilon 0$	T7 item 1: the exact
(SP)	$\tfrac{1}{2} - \tau 0 + \gamma\mu/2 < \tau 0 - \varepsilon 0$	T7 item 3: small-prime trial division, totalled over all trials ("small-prime-total")
(SK)	$r + \tfrac{1}{2} - \tau 0 < \tau 0 - \varepsilon 0$	T7 item 4: sorting the right keys by residue, totalled over all trials ("right-key-sort-total")
(CAP)	$1/\text{guard} + \zeta < \varepsilon 0$	Theorem A′: the cap is reached with probability o(1) ("cap-margin")
(A4G)	$\mu(h(2a') + 2a') + 1/\text{guard} < \eta$	T4: the union bound is o(1) under the guard ("A4R-after-guard")
(PP)	$\tau 0 < \tfrac{1}{2}$	Base Proposition P (l.281–297): the preliminary threshold is separated from its cap
(PD)	$\eta/2 < \tau 0 + 20\varepsilon$	Base l.801: trial division for the large prime

ε0 := least slack of (C1R)–(C6), (MU1), (MU2), (A4R) including its domain condition; at the certified point it is the (C1R) slack, $9.994 \cdot 10^{-7}$. v3.1 requires (EM), (SP) and (SK) with the explicit ε0 slack needed by T7, as in the paper appendix. The frozen point already satisfies these stronger requirements; no input parameter is changed. The ζ term from the trial count is retained in T7's overall exponent. (EM), (PP), (PD) were implicit in v2.

Checksum conditions (paper_v2 Section 8, Base Section 6 with J = 1): Δ > 0, Γ > 0; a_Γ ≤ ¾; p, q ≥ 1; ρ_c < ρ_f; ρ_f + (18 + 6p)ε < η and ρ_f + (18 + 6q)ε < η (Lemma B3's decay rate, which replaces Base's fixed −0.00017); log₂ M1 < ρ_c and log₂ M2 < ρ_c. On a_Γ ≤ ¾: in Lemma B2 (Base l.696–717) the row a_Γ w_v − u_v of the certificate has rounding error (4a_Γ + 3)ε by (6.1) (l.694: $|U - \bar{u}| \le 3\varepsilon$, $|W - \bar{w}| \le 4\varepsilon$), and the accounting 6εΣ_t z_t of the lemma needs 4a_Γ + 3 ≤ 6, i.e. a_Γ ≤ ¾. Base l.702 justifies the 6ε by "coefficients of absolute value at most 1 (a_Γ < 1)", which alone gives only 7ε; the detailed list on l.703 is correct and is what the lemma uses. Here a_Γ = 0.06248, slack 0.6875. The J = 1 certificates: region ρ_0 has z = (p, 0) with exponent p(Δ − u); the bounded regions ρ_1–ρ_8 have z = (q, 0, 0) and the unbounded region ρ_9 has the two-entry vector z = (q, 0) (its G has one band row, l.640), all with exponent q(a_Γ w − u); so the ten certificates are the two integrals M1 and M2. Base's (S1)–(S4), (C7), (C8) and Lemma A4's δ are **not used**.

T.2 The procedure

As Base l.332–383 with these changes.

- **Step 3 (trial count).** Compute |Σ(M)| exactly by the merge procedure of Proposition 2.4 (Base l.1015–1022), without a cap; cost 2^{m − d(M)} poly ≤ 2^{μn} poly. Fix once and for all a polynomial C_S with nonnegative integer coefficients and C_S(m) ≥ max(1, C_S(m)) for every m, where C_S is the polynomial of [S]; so C_S(m) ≤ n^{c_S} for a constant c_S. Put T := ⌈ C_S(m) · n⁸ · ⌈2^m / |Σ(M)|⌉ ⌉. The inner ceiling is of the exact rational 2^m/|Σ(M)| = 2^{d(M)} (d(M) itself is never computed); the outer ceiling is redundant when C_S has integer coefficients and is kept so that T is an integer for any majorant. Since $2^{d(M)} \le \lceil 2^{d(M)} \rceil < 2^{d(M)} + 1 \le 2 \cdot 2^{d(M)}$, C_S(m) n⁸ 2^{d(M)} ≤ T ≤ 4 C_S(m) n⁸ 2^{d(M)}. (Base's sampled F, l.349–353, is not used.)
- **Step 4(c) (left records).** All triples (κ_L, X⁺, X⁻) with κ_L a key of L, X⁺, X⁻ ⊆ M disjoint, |X⁺| = k1, |X⁻| = k2, and b(κ_L) + b^F(X⁺) − b^F(X⁻) ≡ j′ (mod P′).
- **Step 4(d) (right classes).** All Y ⊆ M with |Y| = σ − k1 + k2, grouped by λ(Y) = (b^F(Y), c^F(Y)).
- **Step 4(f) (matching).** For a left record, the group with b(κ_R) + b_λ = T^F − b(κ_L) − b^F(X⁺) + b^F(X⁻), skipping the interval c(κ_R) + c_λ = t^F − c(κ_L) − c^F(X⁺) + c^F(X⁻).
- **Step 4(g) (expansion).** For each visited (κ_R, λ) and member Y: test X⁺ ∩ Y = ∅ and X⁻ ⊆ Y. If both hold, put Z := ((X⁺ ∪ Y) \ X⁻) △ F and insert (κ_L, κ_R, Z) into $\mathscr{D}$_s unless present.
- **Step 6 (cap).** As Base l.379–382, with the count of steps including the exact |Σ(M)| computation. The cap is a property of the real run; Lemmas T5–T7 below are statements about

the *uncapped ideal execution*, the run with the cap removed, and the real run's early stop at the cap is charged separately in Theorem A′.

T.3 Lemmas

Lemma T1 (soundness; Base Lemma A1, l.398–405). Every key inserted into $\mathscr{D}$_s is an alias class of t_s.

Proof. Let $X^+ \cap Y = \varnothing$ and $X^- \subseteq Y$, $Z = ((X^+ \cup Y) \setminus X^-) \triangle F$. Since the sum over a disjoint union is additive and $X^- \subseteq Y$, $b\hat{}F(X^+) - b\hat{}F(X^-) + b\hat{}F(Y) = b\hat{}F((X^+ \cup Y) \setminus X^-) = b(Z) - b(F)$ by (5.1) (Base l.338), and likewise for c. The group condition then gives $b(A) + b(Z) - b(F) + b(B) = T\hat{}F$, so $b(I) = T_{s,q}$ for $I = A \cup Z \cup B$, and the skipped interval gives $c(I) \neq t_s$. The rest is Base's argument. ∎

Lemma T2 (counts; Base Lemma A2, l.411–422). Unchanged: L has at most $(n+1)2\hat{}\{|L| - d(L)\}$ keys; every trial has at most $(2m+1)2\hat{}\{m-d(M)\}$ right classes ($c\hat{}F(Y) = c(Y \triangle F) - c(F) \in \Sigma(M) - c(F)$, independent of $|Y|$).

Lemma T3 (trial count and deficiencies). With probability $1 - o(1)$ over M and L: (i) $d(L) \geq d(M) - 2n\hat{}\{0.6\}$; (ii) under (LD), $d(M) \leq (\tfrac{1}{2} - \tau_0 + \zeta)n + n\hat{}\{0.6\}$.

Proof. M is a uniform m-subset of [n] and L a uniform $|L|$-subset; (F3) (Base l.206–209) gives $|d(M) - f(m)| \leq n\hat{}\{0.6\}$ and $|d(L) - f(|L|)| \leq n\hat{}\{0.6\}$ except with probability $4e\hat{}\{-n\hat{}\{0.2\}/2\}$. Since $|L| > m$ by (MU1) and f is nondecreasing, $d(L) \geq f(|L|) - n\hat{}\{0.6\} \geq f(m) - n\hat{}\{0.6\} \geq d(M) - 2n\hat{}\{0.6\}$. For (ii), $m \leq \lfloor n/2 \rfloor$ by (MU2), so $f(m) \leq f(\lfloor n/2 \rfloor) \leq (\tfrac{1}{2} - \tau_0 + \zeta)n$ under (LD) (Base l.295–297). ∎

Let $\mathcal{E}_3$ be this event; it depends only on M and L.

Lemma T4 (distinctions survive reduction; replaces Base Lemma A4, l.468–476). With probability $1 - o(1)$ over P, every $\lambda \in \{-1,0,1\}\hat{}M$ with at most $2k'$ nonzero entries and $\Sigma\lambda_i c_i \neq 0$ has $P \nmid \Sigma\lambda_i c_i$.

Proof. There are at most $\Sigma_{j \leq 2k'} C(m,j)2\hat{}j \leq 2\hat{}\{(h(2a')+2a')m + O(\log n)\}$ such vectors ($2a' \leq \tfrac{1}{2}$). Each nonzero sum has absolute value below $2\hat{}\{b+O(\log n)\}$, so by (F4) (Base l.210–212) with $L' = \lfloor \eta n \rfloor$ it is divisible by P with probability at most $\mathrm{poly}(n)(b + \log n)2\hat{}\{-\eta n\}$. A union bound gives $\mathrm{poly}(n)(b + \log n)2\hat{}\{-(\eta - \mu(h(2a')+2a'))n\}$. Under the guard, $b + \log n \leq 2\hat{}\{n/10^8 + 2\}$, so this is at most $\mathrm{poly}(n)2\hat{}\{-(\eta - \mu(h(2a')+2a') - 1/10^8)n\} = o(1)$ by (A4G). ∎

On this event, for any $U, U' \subseteq M$ with $|U|, |U'| \leq k'$ and any signs ε_i: $\Sigma_U \varepsilon_i c_i \neq \Sigma_{U'} \varepsilon_i c_i$ implies $\Sigma_U \varepsilon_i b_i \neq \Sigma_{U'} \varepsilon_i b_i$ (the difference is such a λ; Base's argument at l.478–482).

Lemma T5 (one trial; replaces Base Lemma A5, l.486–535). Consider the uncapped ideal execution. Assume (LD), $\mathcal{E}_3$, the event of Lemma T4, and [S] with polynomial constant $C_S = C_S(m)$ ($a_{k'}(M; w) \geq 2\hat{}\{-d(M)\}/C_S(m)$ for every signing w of M, $k' \leq 0.11m$). Fix s, an alias $I \in \mathscr{A}_s(P)$ with carry q, and one trial for (s,q). The trial inserts the key of the class of I with probability at least $2\hat{}\{-d(M)\} / (16\, K_0\, n^2\, (m+1)\, C_S) \geq 2\hat{}\{-d(M)\} / (C_S\, n^6)$, where $K_0 \geq 1$ is the absolute constant of (F4) in Base l.511–512, and the second inequality holds whenever $16K_0(m+1) \leq n^4$, in particular (as $m + 1 \leq n$) for all n with $n^3 \geq 16K_0$. The lemma also needs $\lfloor \gamma m \rfloor \geq L_*$ for (F4). Neither threshold depends on I, s, P or M.

Proof. Steps 1–2 as Base (l.492–499): with probability $\geq 1/(m+1)$, $S := (I \cap M) \triangle F$ has $|S| = \sigma$, and then S is a uniform σ-subset of M; on S the trial's signed weights are $c\hat{}I$ ($c\hat{}I_i = c_i$ on $I \cap M$, $-c_i$ off I), and on $M \setminus S$ they are $-c\hat{}I$.

Step 3 (representations). For $X^+ \subseteq S$, $|X^+| = k1$, and $X^- \subseteq M \setminus S$, $|X^-| = k2$, put $U = X^+ \cup X^-$ and $Y = (S \setminus X^+) \cup X^-$. The left record's reduced weight is $b(A) + b\hat{}F(X^+) - b\hat{}F(X^-) = b(A) + b\hat{}I(U)$ (the sign flip off S cancels the -1 entries). So the family of left keys is $\{b(A) + b\hat{}I(U) : U \in \mathcal{F}(S)\}$ with $\mathcal{F}(S) = \{U \subseteq M : |U \cap S| = k1, |U \setminus S| = k2\}$, and $|\mathcal{F}(S)| = C(\sigma,k1)C(\sigma,k2)$.

Step 3′ (many distinct keys). Lemma R (paper_v2 Section 10; proof: for each distinct k'-sum y of $(M, c\hat{}I)$ fix a witness K_y; $Pr_S[|K_y \cap S| = k1] = C(k',k1)C(m-k', \sigma-k1)/C(m,\sigma)$, and $C(m,k')C(k',k1)C(m-k',\sigma-k1) = C(m,\sigma)C(\sigma,k1)C(\sigma,k2)$) gives $E_S[|\{c\hat{}I(U) : U \in \mathcal{F}(S)\}|/(C(\sigma,k1)C(\sigma,k2))] \geq a_\{k'\}(M; c\hat{}I)$. By (F2) (Base l.204–205) $d(M; c\hat{}I) = d(M)$, so [S] gives $\geq 2\hat{}\{-d(M)\}/C_S$. By Lemma T4 the reduced keys $b(A) + b\hat{}I(U)$ are distinct whenever the $c\hat{}I(U)$ are, so $B_S :=$ #distinct left keys satisfies $E_S[B_S/(C(\sigma,k1)C(\sigma,k2))] \geq 2\hat{}\{-d(M)\}/C_S$. (No "bad S" term: Lemma T4 is global.)

Step 4 (the small prime keeps many residues). Base l.509–520 verbatim with $C(\sigma,k)$ replaced by $C(\sigma,k1)C(\sigma,k2)$: the key set lies in $(-nP, nP)$ and is fixed before P′, j′ are drawn; (F4) with $L' = \lfloor \gamma m \rfloor$ gives $E_\{P'\} \Sigma_y z_y^2 \leq B_S + Kon^2 B_S^2 2\hat{}\{-\gamma m\}$; Markov and Cauchy–Schwarz give, with probability $\geq$ ½ over P′, at least $\min(B_S, 2\hat{}\{\gamma m\})/(4Kon^2)$ occupied residues, so a uniform j′ hits an occupied residue with probability $\geq \min(B_S\ 2\hat{}\{-\gamma m\}, 1)/(8Kon^2) \geq B_S/(8Kon^2\ C(\sigma,k1)C(\sigma,k2))$, using $B_S \leq C(\sigma,k1)C(\sigma,k2)$ and $2\hat{}\{\gamma m\} \leq C(\sigma,k1)C(\sigma,k2)$, which (C1R) gives for all large n (the floors in k1, k2 and the Stirling factors cost $O(\log m)$ in the exponent).

Step 5 (a hit is recovered). If $j' \equiv b(A) + b\hat{}I(U)$ for $U = X^+ \cup X^- \in \mathcal{F}(S)$, the left record $(\kappa(A), X^+, X^-)$ is listed. With $Y = (S \setminus X^+) \cup X^-$: $b\hat{}F(Y) = b\hat{}I(S \setminus X^+) - b\hat{}I(X^-)$, so $b(A) + b\hat{}I(U) + b\hat{}F(Y) + b(B) = b(A) + b\hat{}I(S) + b(B) = b(A) + b(I \cap M) - b(F) + b(B) = T\hat{}F$ (by (5.1) with $Z = S$, $S \triangle F = I \cap M$). The class record $(\kappa(B), \lambda(Y))$ is listed and visited (unreduced sum $c(I) - c(F) \neq t\hat{}F$ as I is an alias); $X^+ \cap Y = \varnothing$ and $X^- \subseteq Y$ hold, and $Z = ((X^+ \cup Y) \setminus X^-) \triangle F = S \triangle F = I \cap M$, so the key $(\kappa(A), \kappa(B), I \cap M)$ is inserted. In the uncapped ideal execution every listed record is processed, so this insertion happens.

Step 6. Multiply: $(1/(m+1)) \cdot (½) \cdot E_S[B_S/(8Kon^2C(\sigma,k1)C(\sigma,k2))] \geq 2\hat{}\{-d(M)\}/(16Kon^2(m+1)C_S)$, and $16Kon^2(m+1) \leq n^6$ when $16Ko(m+1) \leq n^4$. ∎

Corollary T6 (coverage; Base Corollary A6, l.537–543). In the uncapped ideal execution, under the assumptions of Lemma T5 and for n above its thresholds, except with probability o(1) every alias class of every target is in its dictionary at the end of Step 4.

Proof. Condition on P, φ, M, L. The T trials for a pair (s,q) are independent, and $T \geq C_S(m)\ n^8\ 2\hat{}\{d(M)\} \geq C_S\ n^8\ 2\hat{}\{d(M)\}$ by Step 3. By Lemma T5 all of them miss a fixed alias class with carry q with probability at most $(1 - 2\hat{}\{-d(M)\}/(C_S\ n^6))\hat{}T \leq \exp(-T \cdot 2\hat{}\{-d(M)\}/(C_S\ n^6)) \leq e\hat{}\{-n^2\}$. (The constant C_S cancels: the trial count carries it. Each target/subset pair has a unique carry, so no factor for carries is needed.) There are at most $(nKo+1)2\hat{}n$ pairs of a target and a subset, and $(nKo+1)2\hat{}n\ e\hat{}\{-n^2\} = o(1)$, with a threshold depending only on Ko. ∎

Lemma T7 (expected work; Base Lemma A7, l.547–589). Assume (LD) and $\mathcal{E}_3$; condition on M, L; average over P and the trials; the phases are arbitrary. The uncapped ideal execution takes in

expectation at most (b+2)2^{(τ0 − ε0 + ζ)n + O(n^{0.6})} steps.

Proof. T ≤ 4C_S(m)n⁸2^{d(M)} ≤ 2^{d(M) + O(log n)} by Step 3; with T3(i), T ≤ 2^{d(L) + 2n^{0.6} + O(log n)}; with T3(ii), T ≤ 2^{(½−τ0+ζ)n + O(n^{0.6})}. There are O(n³) pairs (s,q). Polynomial factors (including C_S(m) ≤ n^{c_S}) go into 2^{O(n^{0.6})}.

1. *Value maps and Step 3:* 2^{|L|} + 2^{|R|} + 2^{m − d(M)}poly: exponents l, r, μ < τ0 − ε0 by (C6), (MU1) and (EM).
2. *Masks:* per trial ≤ 2^{H3(a,c)m + O(log n)} + 2^{h(½−a+c)m + 1} (both lists, every trial); times T ≤ 2^{(½−τ0+ζ)n + O(n^{0.6})}: exponent max(H3, h(½−a+c))μ + ½ − τ0 + ζ < τ0 − ε0 + ζ by (C5R).
3. *Small primes:* n³2^{γm/2+1} divisions per trial; total 2^{(½ − τ0 + ζ + γμ/2)n + O(n^{0.6})}; below the bound by (SP).
4. *Sorting keys by residue:* (n+1)(2^{|L|−d(L)} + 2^{|R|}) per trial; total ≤ 2^{ln + O(n^{0.6})} + 2^{(r + ½ − τ0 + ζ)n + O(n^{0.6})} (T3); below the bound by (C6) and (SK).
5. *Left records:* (n+1)2^{|L|−d(L)} · 2^{H3(a,c)m + O(log n)} · 2^{1−γm} per trial in expectation (for fixed choices other than j′, a triple is listed with probability 1/P′ ≤ 2^{1−γm}); T ≤ 2^{d(L) + 2n^{0.6} + O(log n)} (T3(i)); total 2^{(l + (H3(a,c) − γ)μ)n + O(n^{0.6})}. Use (C2R).
6. *Right class records:* (n+1)2^{|R|}(2m+1)2^{m−d(M)}2^{1−γm} per trial; T ≤ 2^{d(M)+O(log n)} cancels d(M) exactly: 2^{(r + (1−γ)μ)n + O(n^{0.6})}. Use (C3). Listing the classes themselves costs 2^{m + O(log n)}; use (EM).
7. *Visits and expansions:* count every expanded member, including those failing the test X⁺ ∩ Y = ∅, X⁻ ⊆ Y. Fix unreduced values x_L ∈ Σ(L), x_R ∈ Σ(R) and masks X⁺, X⁻, Y before P, and put D = x_L + c^F(X⁺) − c^F(X⁻) + x_R + c^F(Y) − t^F, independent of P. A visited tuple (κ_L, X⁺, X⁻, κ_R, Y) with c(κ_L) = x_L, c(κ_R) = x_R needs D ≠ 0 (the exact interval is skipped), P | D (the reduced weights add to T^F ≡ t^F mod P), and the two small-prime conditions, which given everything else hold for at most one residue j′. By (F4), Pr_P[P | D] ≤ poly(n)(b + log n)2^{−ηn} as |D| < 2^{b+O(log n)}; by Lemma T2 at most (n+1)² key pairs share (x_L, x_R). Per trial ≤ poly(n)(b + log n)2^{|L|−d(L)+|R|−d(R)} · 2^{H3(a,c)m} · 2^{h(½−a+c)m} · 2^{−ηn−γm}; with d(R) ≥ 0 and T ≤ 2^{d(L)+2n^{0.6}+O(log n)}: (b + log n)2^{(1 − μ + (H3(a,c) + h(½−a+c) − γ)μ − η)n + O(n^{0.6})}. Use (C4R). This averages over P without conditioning on the event of Lemma T4; only coverage uses that event.
8. *Matching, dictionaries, output:* polynomial per record, tuple and key. ∎

Theorem A′. Under [S] with polynomial constant C_S(m), the procedure of T.2 has the properties of Base's Theorem A (l.319–330) with the constants of T.1, for all sufficiently large n: on (LD), with P uniform and φ arbitrary, it returns with probability 1 − o(1) values Â_s within 1/16 of Σ_{I∈𝒜_s(P)} χ(I) for every s; on every execution it uses at most 2^{τ0 n + 2n^{0.7}} poly(n+b+2) word operations.

Proof. As Base Section 5.8 (l.593–612). Part 2 is the cap. For part 1, the failure events, combined by a union bound: $\mathcal{E}_3$ fails with probability o(1) (T3); Lemma T4's event fails with probability o(1) (M is independent of P); given both, some alias class is missed in the uncapped ideal execution with probability o(1) (T6); the cap is reached, which by Markov's inequality and Lemma T7 (not

conditioned on T4's event) has probability o(1) + (b+2)2^{(τ0−ε0+ζ)n+O(n^{0.6})}/2^{τ0 n + n^{0.7}} = o(1), because b + 2 ≤ 2^{n/10⁸+1} and 10^{-8} + ζ < ε0 (CAP); a bounded sampler fails with probability 2^{O(n)}(2^{−n³} + e^{−con²}) = o(1). Outside these events the real run coincides with the uncapped ideal execution, $\mathscr{D}$_s contains every alias class (T6) and nothing else (T1), and the output precision is as in Base (l.608–612). ∎

Scope. With Theorem A′ in place of Theorem A, Base's Theorem 1 (l.765–810) gives, in Base's word-RAM model (word length (1.1) of the paper), a uniform randomized algorithm that halts on every execution, answers correctly with probability at least 2/3 for every input with n ≥ 2 (exact fallback below the cutoff n_* and when the guard fails), and on every execution with n ≥ n_* that passes the guard uses at most 2^{(τ0 + 20ε)n + 2n^{0.7}} poly(n + b + 2) word operations. For each fixed positive integer c, every input with n ≥ N_c and b ≤ n^c passes the guard, and the factor 2^{2n^{0.7}}poly(n + n^c + 2) is absorbed by any fixed T′ > τ0 + 20ε; the address-space argument of Base l.809–810 applies unchanged. This gives O_{c,T′}(2^{T′n}) word operations for every T′ > T := τ0 + 20ε. The pure exponential bound is claimed only in that range (b ≤ n^c for a fixed c); for larger b, only the displayed bound with its poly(n + b + 2) factor and the exact fallback are claimed, as in Base l.58–66 and 806–810.

T.4 The certified point (`candidate_ternary.json`)

Parameter	Value
τ0	0.4873405528
η	0.5159974155
μ	0.4746795055
a, c	0.0035141538, 0.0035141537 (a′ = 0.0070283075)
γ	0.0603737099
l	0.4840025845
κ, p, q_m	1.03462910, 8.46825883, 8.27946618
filter	16 cosine + 16 sine terms per density, 8 decimals (in the JSON)
ε, ζ, guard	10^{-8}, 10^{-8}, 10^{8}
ρ_f, ρ_c	η − $1.5 \cdot 10^{-6}$, ρ_f − 10^{-6}
T = τ0 + 20ε	**0.4873407528**

All 33 conditions of `scripts/round2/check_ternary_point.py` pass in arb at 200 bits (`scripts/round2/out/ternary_certificate_N4096_8192.txt`, re-run unchanged in round 7). Tightest: C1R $9.99 \cdot 10^{-7}$ (= ε0), B3 $8.1 \cdot 10^{-7}$, C2R/C3/C4R $1.30 \cdot 10^{-6}$, C5R $1.60 \cdot 10^{-6}$, M1 $3.04 \cdot 10^{-6}$, M2 $3.30 \cdot 10^{-6}$. The conditions added to the table in v3 have slacks (SP) 0.4604, (SK) 0.4334, (CAP) $9.79 \cdot 10^{-7}$, (A4G) 0.4587, (EM) 0.01266, (PP) 0.01266, (PD) 0.2293 (`scripts/round7/out/recheck_v3.txt`); they agree with Astra's independent 50-digit intervals (`SCALAR_SLACKS.md`), which also certify both moments with an independently written interval FFT: $\log_2$ M1 = 0.51599187405865778648, $\log_2$ M2 = 0.51599162047014794189, ball radii below $4 \cdot 10^{-55}$,

identical at N = 1024, 4096, 8192 here and at N = 4096, 8192 in Astra's certificate. The point is $2.3465 \cdot 10^{-4}$ below the certified Section 8 point (T = 0.4875754036) and $3.352 \cdot 10^{-4}$ below JoshuaQFEX.

Statement. *Conditional on [S] and on Base's written proof*: there is a uniform randomized algorithm for Subset Sum in Base's word-RAM model that halts on every execution and answers correctly with probability at least 2/3 on every input with n ≥ 2, and, for every fixed positive integer c and every fixed T′ > T = 0.4873407528, processes every input with n ≥ N_{c,T′} and b ≤ n^c in at most C_{c,T′} · 2^{T′n} word operations on every outcome of its random choices; in particular O_c(2^{0.4873408n}) for b ≤ n^c. (Not 0.48734: that is below T. The threshold is the exact decimal T = τ0 + 20ε; no exponent is rounded down.) For inputs with b above every fixed polynomial in n, only the bound 2^{Tn + 2n^{0.7}}poly(n + b + 2) under the guard, and the exact fallback otherwise, are claimed.

Cutoffs. The o(1) terms depend only on n (and on the fixed constants of Base and of [S]), so a fixed cutoff n_* exists; it is not numeric, and several thresholds enter it:

- Base's preliminary search: $\zeta = 10^{-8}$ makes Proposition P need $n > 10^8$ and its Chebyshev term $4/(\zeta^2 n)$ small only for n of order 10^{17} (as for the Section 8 point).
- Lemma T5: (F4) needs ⌊γm⌋ ≥ L_*, and the bound 2^{−d(M)}/(C_S n^6) needs 16Ko(m+1) ≤ n^4, which holds for n^3 ≥ 16Ko since m + 1 ≤ n; Ko and L_* are the absolute constants of (F4). The polynomial C_S(m) of [S] is absorbed into the trial count (Step 3) and into 2^{O(n^{0.6})} (Lemma T7), and cancels in Corollary T6, so no threshold depends on C_S; but the thresholds above remain. (The analogous cutoff for Base's own Lemma A5 is $32Kon^2(m+1)(v+1) \le n^6$, a different inequality; it is not used here.)
- Corollary T6: (nNo+1)2^n e^{−n²} ≤ 1/n, a threshold depending on No.
- Theorem A′: the cap probability (b+2)2^{(τ0−ε0+ζ)n+O(n^{0.6})−τ0n−n^{0.7}} is o(1) once the exponent $-(\varepsilon 0 - \zeta - 10^{-8})n - n^{0.7} + O(n^{0.6}) + 1$ is negative; (CAP) gives $\varepsilon 0 - \zeta - 10^{-8} = 9.79 \cdot 10^{-7} > 0$, and the threshold depends on the O(n^{0.6}) constants of Lemma T7 and (F3). Separately, the absorption of 2^{2n^{0.7}}poly(n + n^c) into 2^{(T′−T)n} has its own threshold N_{c,T′}. Exact layer-density monotonicity is false (ROUND2_REPORT A.3, m = 8), so [S] is taken from Conjecture M′ (a_t ≤ P(m) · a_k for t ∈ [k, m−k], k ≤ 0.11m) through Lemma S, which gives C_S(m) = P(m) + 2.